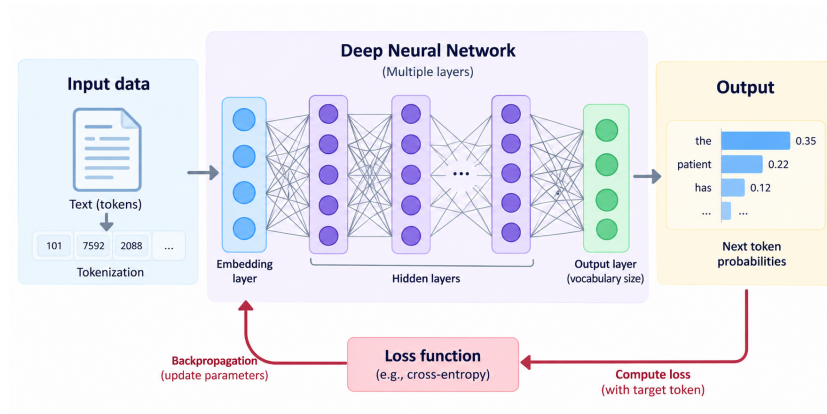




Deep Learning Study

Medical Domain Adaptation of Llama Models

Fine-tuning, Layer Freezing, and Knowledge Distillation



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Medical Domain Adaptation of Llama Models

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This study investigates transfer-learning strategies for adapting pre-trained Llama models to medical-domain text under limited GPU memory. The evaluated approaches are parameter-efficient fine-tuning, selective layer freezing, and offline knowledge distillation. In the freezing experiment, only the final two transformer layers of Llama-3.1-8B-Instruct were updated. In the distillation experiment, an 8B teacher transferred knowledge to a LoRA-adapted Llama-3.2-1B-Instruct student through a temperature-scaled top- K objective. Distillation reduced validation perplexity from 26.51 to 22.54. These preliminary results illustrate the trade-offs among model quality, memory efficiency, and training cost, but the small subsets and single-run evaluation limit the strength of the conclusions.

1. Project Overview

This study investigates how pretrained language models can be adapted to a specialized medical domain. We evaluate three transfer-learning strategies: parameter-efficient fine-tuning, selective layer freezing, and offline knowledge distillation. The main objective is to compare their effectiveness and resource requirements.

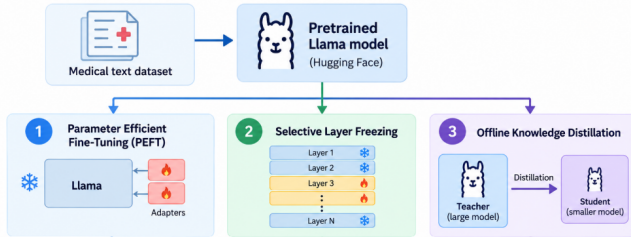


Fig. 1. Overview of the three transfer-learning strategies evaluated for medical domain adaptation.

The study focuses on causal language modeling, in which models learn to predict the next token in medical text. It does not address clinical diagnosis or treatment recommendations. Fine-tuning supports domain adaptation, while layer freezing reduces the number of trainable parameters and therefore the computational workload. Distillation transfers knowledge from a larger teacher model to a smaller student. Performance is evaluated using validation loss and perplexity, with lower values indicating better predictive performance.

2. Method

A. Overall Strategies. The study compares three approaches for adapting pretrained models to a specialized medical domain under limited computational resources. First, each pretrained model is evaluated on medical text to establish baseline causal language-modeling loss and perplexity. The selected adaptation strategy is then applied under memory-efficient settings, including 4-bit quantization. Fine-tuning updates a parameter-efficient subset of model weights, layer freezing updates only the final two transformer layers, and offline distillation trains a smaller student to reproduce information from a larger teacher.

This design evaluates the trade-off between domain-adaptation quality and training efficiency rather than focusing only on the highest possible performance.

B. Models Used. The primary model is Meta Llama-3.1-8B-Instruct, an instruction-tuned causal language model with eight billion parameters. It serves as the base model for fine-tuning and layer freezing and as the teacher for distillation. Meta Llama-3.2-1B-Instruct is the student model. The student is adapted with LoRA, allowing it to learn from the teacher while updating only a small set of parameters. The models are loaded from Hugging Face with 4-bit quantization to reduce GPU memory consumption during training and inference.

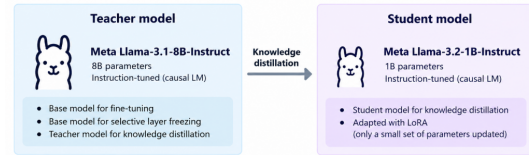


Fig. 2. Teacher and student models used in the adaptation experiments.

C. Dataset and Exploratory Analysis. The selected dataset was the Saminx22 medical-data-for-SLM dataset from Hugging Face. It contains medical text extracted from PubMed, with 42,187 training documents and 2,221 validation documents. The exploratory analysis confirmed that the dataset contains no missing or empty text records and no duplicate training documents. All records originate from PubMed. Documents are relatively long, with an average of 2,704 words and a median of 2,317 words. Tokenization of a 2,000 document sample produced an average length of 3,417 Llama tokens. The dataset also showed a large vocabulary, approximately 395,445 unique non-stopwords, reflecting the specialized and diverse terminology of biomedical literature.

D. Data Processing. The data were checked for missing values, empty records, duplicate texts, and extreme document lengths. Exploratory analysis examined word counts, character counts, vocabulary characteristics, and the distribution of text lengths. Texts were tokenized with the corresponding Llama tokenizer. Sequences were truncated and padded to a maximum length of 512 tokens to provide consistent input sizes and control GPU memory usage. The freezing experiment used 1,500 training documents and 200 validation documents. The distillation feasibility experiment used 100 training documents and 50 validation documents.

The knowledge distillation objective combines the standard causal language modeling loss with a distillation loss that encourages the student model to reproduce the teacher model's output distribution:

$$\mathcal{L}_{KD} = \alpha \mathcal{L}_{CE} + (1 - \alpha) T^2 D_{KL} \left(P_T^{(T)} \parallel P_S^{(T)} \right), \quad [1]$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss, $P_T^{(T)}$ and $P_S^{(T)}$ are the temperature-scaled output distributions of the teacher and student models, respectively, T is the temperature parameter, and α controls the balance between learning from the ground-truth tokens and imitating the teacher.

$$\theta_i^{(t+1)} = \begin{cases} \theta_i^{(t)} - \eta \nabla_{\theta_i} \mathcal{L}, & \theta_i \in \Theta_{\text{trainable}}, \\ \theta_i^{(t)}, & \theta_i \in \Theta_{\text{frozen}}, \end{cases} \quad [2]$$

where θ_i represents a model parameter, η is the learning rate, and $\mathcal{L}$ is the causal language modeling loss. Parameters belonging to frozen layers remain unchanged, reducing the number of parameters that require gradient updates during training.

The experiments used predefined training and validation splits. Model quality was evaluated on validation text using causal-language-modeling loss and perplexity. Lower values indicate better prediction. The freezing and distillation experiments used medical data, while the supplied fine-tuning configuration uses a separate KYC dataset.

Table 1. Compact training setup.

Method	Data	Key setup and metrics
Fine-tuning	60 KYC training examples	Llama-3.1-8B; QLoRA; 3 epochs; LR 2×10^{-4} ; ROUGE-L and semantic similarity.
Freezing	1,500 train / 200 validation	Llama-3.1-8B; final 2 layers unfrozen; 1 epoch; LR 2×10^{-5} ; loss and perplexity.
Distillation	100 train / 50 validation	8B teacher, 1B student; LoRA; LR 2×10^{-4} ; $T = 2$; loss and perplexity.

The freezing baseline used a validation perplexity of 8.66 before adaptation. For distillation, the reported student validation perplexity decreased from 26.51 before adaptation to 22.54 after distillation. Because the experiments use different models and validation subsets, these values should be interpreted within each experiment rather than compared directly across all three strategies.

3. Results

The strongest quantitative result was obtained from the knowledge-distillation experiment. The Llama-3.2-1B student achieved a validation loss of 3.1154 and a perplexity of 22.54 after distillation. Compared with the reported unadapted student baseline (loss: 3.2776; perplexity: 26.51), this corresponds to a 4.9% reduction in validation loss and a 15% reduction in perplexity.

Table 2. Validation results recorded in the experiments. Lower values are better.

Method	Base loss	Base PPL	Adapted loss	Adapted PPL
Distillation	3.2776	26.51	3.1154	22.54
Freezing	2.1591	8.66	—	—

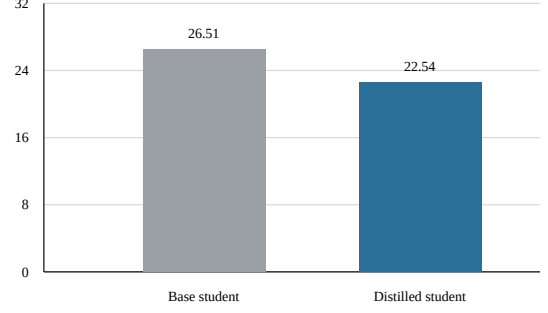


Fig. 3. Validation perplexity of the Llama-3.2-1B student before and after knowledge distillation.

For the layer-freezing experiment, the pretrained Llama-3.1-8B model obtained a baseline perplexity of 8.66 on the 200-document validation subset before adaptation. The training loss curve fluctuated between 1.76 and 2.09 across the training epoch.

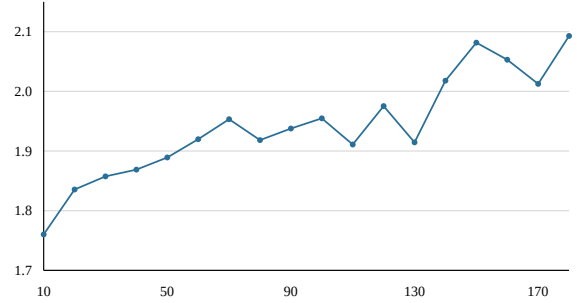


Fig. 4. Training loss trajectory during the layer-freezing experiment.

The fine-tuning experiment used different KYC data and evaluation metrics, including ROUGE-L, semantic similarity, and keyword recall. Overall, the results provide evidence that offline knowledge distillation can improve a compact medical language model, although the experiments are not directly comparable because they use different datasets, models, and sample sizes.

4. Discussion

The results suggest that offline knowledge distillation is a promising approach for adapting compact language models to medical text. The distilled Llama-3.2-1B student reduced validation perplexity from 26.51 to 22.54, showing that knowledge from the larger Llama-3.1-8B teacher improved next-token prediction.

The experiments also demonstrate that 4-bit quantization and parameter-efficient adaptation make Llama-based training feasible on a single GPU. However, the freezing experiment did not retain a post-training validation score, so its effectiveness cannot be evaluated reliably.

The distillation experiment used only 100 training documents and 50 validation documents, with one training run and no repeated seeds.

The fine-tuning experiment used a KYC dataset and different evaluation metrics, making it unsuitable for direct comparison with the medical experiments.

Future work should evaluate all strategies on the same medical split, repeat training with multiple random seeds, record post-training metrics consistently and use larger training subsets. Finally, these results measure language-modeling quality only and do not demonstrate clinical accuracy or safety; the models should not be used for diagnosis or treatment decisions.